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Department of Artificial Intelligence and Machine Learning

Date: 7.11.2025	Test -1	Max. Marks: 50+10
Semester: V	UG	Duration: $1\frac{1}{2}$ Hrs + ½ Hr
Course Title: Artificial Neural Networks and Deep Learning		Course Code: AI253IA

Part A

Sl No	Questions	M	BT	CO
1.	A perceptron has weights $w_1 = 0.4$, $w_2 = and\ -0.2$, $b = 0.1$. For input $x_1 = 1$, $x_2 = 0$, compute the net input and output (step activation).	1	3	1
2.	Compute the output of a sigmoid neuron if $net = 0.8$.	1	2	1
3.	Define learning rate. Why is it important in neural network training?	1	2	2
4.	Given inputs $x_1 = 1$, $x_2 = 2$, desired output $d = 8$, initial weights $w = [0.5, 1.0]$, and $\eta = 0.1$. Compute updated weights using the LMS rule after one iteration.	2	4	2
5.	Mention two advantages of using multilayer perceptrons over single-layer perceptrons.	2	3	1
6.	Compute the total number of weights in a 3-2-1 MLP (3 inputs, 2 hidden, 1 output; include bias).	2	3	1
7.	What are the sequential and batch modes of training?	1	2	2

Part B

Sl No	Questions	M	BT	CO
1a.	Explain the structure and mathematical model of an artificial neuron. Discuss how the activation function affects the neuron's decision boundary.	04	2	1
1b.	A neuron receives three inputs: $x_1 = 1$, $x_2 = 0$, $x_3 = 1$ with weights $w_1 = 0.2$, $w_2 = -0.3$, $w_3 = 0.4$, and bias $b = 0.1$. i. Compute the net input and the output using a sigmoid activation. ii. Recompute output if $b = 0.5$and comment on its effect.	06	3	1
2a.	Compare and contrast Hebbian, Competitive, and Error-Correction learning. Give one practical example where each learning rule can be applied.	05	3	2
2b	Given an adaptive filter with $x = [1,2]$, $w = [0.5, -0.5]$, $d = 3$, and $\eta = 0.05$, compute: 1. Output $y = w^T x$ 2. Error $e = d - y$ 3. Updated weights $w_{new} = w + \eta e x$.	05	4	2
3	Explain the Perceptron Learning Algorithm with its update rule and convergence theorem. Why does the perceptron fail for non-linearly separable problems?	06	2	2
4	The network weights have been initialized as shown in Fig.1. Analyze the Multilayer perceptron Back Propagation algorithm for the initialized network by doing the following (use the sigmoid function): with a learning rate $\eta = 0.25$. i. Compute the net input and output of each hidden neuron and the final output neuron using the forward pass.	14	4	3

ii. Calculate the error term, the delta values for both output and hidden neurons, and determine the gradient direction.

iii. Update the weights using the backpropagation learning rule and show how the error decreases over two iterations of training.

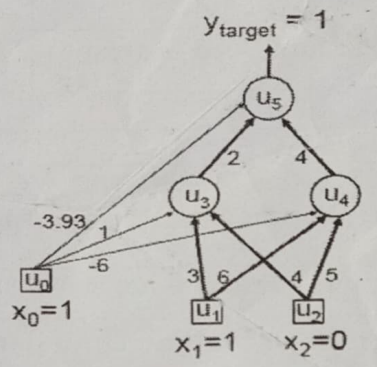

Fig. 1.

5 Consider the single-neuron perceptron network shown in Figure 2 below. The network is used to classify vehicles as either a BUS (+1) or a CAR (0) based on three input parameters: x_1: Weight, x_2: Fuel Capacity, x_3: Number of Passengers. A bias input $x_0 = 1$ is also provided.

Tbest = 0.5

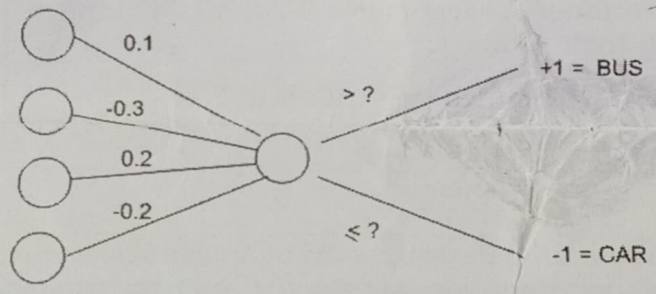

Fig. 2.

Given: Learning rate, $\eta = 0.1$, Activation: Sigmoid, Input vector (with bias): $\mathbf{x} = (1, 10, 12, 13)$, Target output: $t = 1$(**BUS**) | 10 | 4 | 2

Tasks

i. Compute the net input and the sigmoid output $y = f(net)$for the given input. State whether the perceptron currently classifies it as a car or a bus.

ii. Suppose this input was misclassified. Using the sigmoid delta learning rule, compute the weight corrections:

$$\Delta w_i = \eta(t - y)\, y(1 - y)\, x_i \qquad \text{for } i = 0,1,2,3$$

iii. Determine the new weights after this update.

iv. Using the updated weights, calculate the new output y' for the same input. Does the network now correctly classify the example?

v. A second input $(1, 5, 7, 7)$ was previously correctly classified as a car. After weight updates, discuss briefly whether this example may now be misclassified (over-correction).